Marking Scheme

M1-SET-1 · M1 · 100 marks

FOR REVISION

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating： **4.0** / 5.0

General Rules
• M: method mark (awarded for a correct method; follow-through allowed)
• A: accuracy mark (depends on the related M mark; no mark for a wrong answer)
• Unmarked: independent mark • f.t.： follow-through • r.t.： accept rounded answer
• M：方法分，可 follow-through；A：答案分，須先得相關 M 分；B：獨立分。
• r.t.：接受四捨五入至四位小數的答案；f.t.：結論按考生前面答案跟進給分。
• 後續小題應使用未四捨五入的中間數值；標準正態分佈表的 z 值取兩位小數。
• 其他正確方法亦給全分。
• 正態分佈題以查表值為準；用計數機計算 Φ 所得答案（例如 0.0899、0.5129 一類差異）亦接受。
• Other correct methods should also be given credit.

Section A (50 marks)

1. Discrete random variables

a (undefined mark)

M1 $k(5c-15)=1$ and $k(15c-55)=\frac{7}{3}$
兩條方程都要列出

A1 $c=6$

A1 $k=\frac{1}{15}$

Common error：漏咗用 $\sum P(X=x)=1$，只得一條方程。

b (undefined mark)

M1 $E(X^2)=\frac{1}{15}\sum x^2(6-x)=7$, $\text{Var}(X)=7-\left(\frac{7}{3}\right)^2=\frac{14}{9}$

A1 $\text{Var}(\bar{X})=\frac{14/9}{56}=\frac{1}{36}$

M1 $\bar{X}\sim N\left(\frac{7}{3},\frac{1}{36}\right)$ approx.; $P\left(Z>\frac{2.5-7/3}{1/6}\right)=P(Z>1)$
分佈陳述可被吸收

A1 0.1587
r.t. 0.1587

Common error：用 $\text{Var}(X)$ 而唔係 $\text{Var}(X)/56$ 標準化。

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2. Binomial distribution

a (undefined mark)

M1 $C_5^6p^5(1-p)=2p^6\Rightarrow 6(1-p)=2p$

A1 $p=\frac{3}{4}$

b (undefined mark)

M1 $E(Y)=6\times\frac{3}{4}=4.5$

M1 $P(Y>4.5)=P(Y=5)+P(Y=6)=C_5^6\left(\frac{3}{4}\right)^5\left(\frac{1}{4}\right)+\left(\frac{3}{4}\right)^6$

A1 $\frac{2187}{4096}\approx 0.5339$
r.t. 0.5339

Common error：誤以為 $Y>4.5$ 包括 $Y=4$。

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3. Confidence intervals

a (undefined mark)

M1 $\bar{x} = \frac{1632}{64}$; $s = \sqrt{\frac{1}{63}(41978.88 - 64(25.5)^2)}$

A1 $\bar{x} = 25.5$, $s = 2.4$
兩個都要啱

b(i) (undefined mark)

M1 95% CI $= \left(25.5 - 1.96 \times \frac{2.4}{8},\ 25.5 + 1.96 \times \frac{2.4}{8}\right) = (24.912,\ 26.088)$

A1 It contains 25, so disagree.
f.t.

b(ii) (undefined mark)

M1 $25.5 - 1.96 \times \frac{2.4}{\sqrt{m}} > 25$

M1 $\sqrt{m} > 9.408 \Rightarrow m > 88.51$

A1 $m = 89$

Common error：把 $m > 88.51$ 四捨五入做 *88*（*m* = *88* 時下限為 *24.9986*，仍包含 *25*）。

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4. Probability

a (undefined mark)

M1 $P(B) = \frac{0.12}{0.3} = 0.4$

M1 $P(A \cup B) = 1 - P(A' \cap B') = 0.6$; $P(A) = 0.6 - 0.4 + 0.12$

A1 $P(A) = 0.32$

b (undefined mark)

$P(A)P(B) = 0.128 \neq 0.12 = P(A \cap B)$, so A and B are not independent.
須有比較

c (undefined mark)

M1 $0.32 + P(C) - 0.32P(C) = 0.66$

A1 $P(C) = 0.5$

Common error：用了互斥公式 $P(A) + P(C) = 0.66$。

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5. Differentiation

a (undefined mark)

M1 $f'(x) = 2e^{2x} - 6e^{x} + 4$

A1 $= 2\left(e^{2x} - 3e^{x} + 2\right) = 2(e^{x} - 1)(e^{x} - 2)$
給定結果，須有因式分解步驟

b (undefined mark)

M1 For $0

A1 so $e^{x} - 1 > 0$ and $e^{x} - 2 < 0$, hence $f'(x) < 0$ (e.g. $f'(0.3) \approx -0.4549$)

A1 $f(x)$ decreases for $0 *f.t.*

Alternative：只代一個 $0

Common error：只睇 x 很大時 $f'(x) > 0$ 就同意。

6. Binomial expansion

a (undefined mark)

M1 $k^{x} = e^{x \ln k}$

A1 $k^{x} = 1 + (\ln k)x + \frac{(\ln k)^2}{2}x^2 + \cdots$

b(i) (undefined mark)

M1 Coefficient of x^2: $6 + 4\ln k + \frac{(\ln k)^2}{2} = 0 \Rightarrow (\ln k)^2 + 8\ln k + 12 = 0$

M1 $\ln k = -2$ or -6; coefficient of x is $4 + \ln k > 0$, so reject $\ln k = -6$

A1 $k = e^{-2}$

Common error：忘記用「x 的係數為正數」捨去一個根。

b(ii) (undefined mark)

A1 $0 \times 1 + (4 - 2) \times 1 = 2$

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7. Definite integrals and areas

a (undefined mark)

M1 $\frac{d}{dx}\left[(x^2 + 1)e^{-x^2}\right] = 2xe^{-x^2} - 2x(x^2 + 1)e^{-x^2}$

A1 $= -2x^3e^{-x^2}$

A1 $\int x^3e^{-x^2}\,dx = -\frac{1}{2}(x^2 + 1)e^{-x^2} + 常數$
漏常數扣此分

b (undefined mark)

M1 Total area $= -\int_{-1}^{0} x^3e^{-x^2}dx + \int_{0}^{1} x^3e^{-x^2}dx$
須處理 $x < 0$ 段在 x 軸下方

M1 $= 2\left[-\frac{1}{2}(x^2 + 1)e^{-x^2}\right]_0^1$

A1 $1 - \frac{2}{e} \approx 0.2642$

Common error：直接計 $\int_{-1}^{1} x^3e^{-x^2}dx = 0$。

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8. Applications of differentiation

a (undefined mark)

M1 $\frac{dy}{dx} = a - \frac{16}{(x+1)^3}$; $a - 2 = 0$ and $a + b + 2 = 5$

A1 $a = 2, b = 1$

b(i) (undefined mark)

M1 $\frac{dy}{dx} = 0 \iff (x+1)^3 = 8 \iff x = 1$ only; $\frac{d^2y}{dx^2} = \frac{48}{(x+1)^4} > 0$
或用一階導數符號測試

A1 $(1, 5)$ is a minimum point

A1 Only one turning point, which is a minimum, so Γ has no maximum point.
f.t.

Common error：只計 y'' 而冇說明 $y' = 0$ 只得一個解。

b(ii) (undefined mark)

M1 At $x = 3$: $y = 6 + 1 + \frac{8}{16} = 7.5$

M1 $\frac{dy}{dx} = 2 - \frac{16}{64} = \frac{7}{4}$

A1 $7x - 4y + 9 = 0$
或 $y = \frac{7}{4}x + \frac{9}{4}$

Section B (50 marks)

9. Poisson distribution

a (undefined mark)

A1 $\frac{\lambda^2 e^{-\lambda}}{2} = 1.2\lambda e^{-\lambda} \Rightarrow \lambda = 2.4$

b (undefined mark)

M1 Mean for 20 minutes $= 2.4 \times \frac{1}{3} = 0.8$; $1 - e^{-0.8} - 0.8e^{-0.8}$

A1 ≈ 0.1912
r.t. 0.1912

c(i) (undefined mark)

M1 Mean for 3 hours $= 7.2$; $\frac{e^{-7.2}(7.2)^4}{4!}$

A1 ≈ 0.0836
r.t. 0.0836

c(ii) (undefined mark)

M1 Fruitful with total 4 $\iff$ counts are a permutation of $(2, 1, 1)$: $3 \times \frac{e^{-2.4}2.4^2}{2!}\left(e^{-2.4}2.4\right)^2$

M1 Divide by the answer of (c)(i)

A1 $\frac{4}{9} \approx 0.4444$
r.t. 0.4444

Alternative：已知共 *4* 顆，每顆流星獨立地以 $\frac{1}{3}$ 的概率落在每一小時：$3 \times \frac{4!}{2!1!1!}\left(\frac{1}{3}\right)^4 = \frac{4}{9}$，全分。
Common error：漏乘排列數 *3*。

c(iii) (undefined mark)

M1 $p = (1 - e^{-2.4})^3 \approx 0.7518$, $P_n = C_{n-1}^{n} p^{n-1}(1-p) = np^{n-1}(1-p)$

M1 $\frac{P_{n+1}}{P_n} = \frac{(n+1)p}{n}$
或比較相鄰項

M1 $\frac{P_{n+1}}{P_n} > 1 \iff n < \frac{p}{1-p} \approx 3.0288$, so $P_1P_5>P_6>\cdots$
或 $P_3 \approx 0.4209 < P_4 \approx 0.4219 > P_5 \approx 0.3964$ 並說明之後遞減

A1 Agree
f.t.

Common error：只比較 P_3、P_4、P_5 而冇說明 $n \geq 5$ 時一直遞減；或 p 取兩位小數（*0.75*）令 $P_3 = P_4$。

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10. Normal distribution

a (undefined mark)

M1 $\frac{26-\mu}{\sigma} = 1.2$ and $\frac{18-\mu}{\sigma} = -2$

A1 $\mu = 23, \sigma = 2.5$

b (undefined mark)

M1 $P(T > 25) = P(Z > 0.8) = 0.2119$; $P(T < 20) = P(Z < -1.2) = 0.1151$

M1 $\frac{P(T>25)}{1-P(T<20)} = \frac{0.2119}{0.8849}$

A1 ≈ 0.2395
r.t. 0.2395

c(i) (undefined mark)

A1 **0.1151**
以查表為準；用計數機算 Φ 的答案亦接受

c(ii) (undefined mark)

M1 $First7phonesallnotdefective$

A1 $(0.8849)^7 \approx 0.4249$
r.t. 0.4249

Common error：寫成 0.8849^8（第 *8* 部無論如何都會被檢查）。

c(iii1) (undefined mark)

M1 Expected number $= 7q + 0.2119$

A1 ≈ 1.8881
r.t. 1.8881

A1 $1.8881 > 1.8$, so yes.
f.t.

Common error：用 $8 \times 0.2119 = 1.6952$ 而得出不同意。

c(iii2) (undefined mark)

M1 First 7: each durable with probability $q = \frac{0.2119}{0.8849}$ (from (b)); 8th: durable with probability 0.2119

M1 $C_2^7 q^2 (1-q)^5 (0.7881) + C_1^7 q (1-q)^6 (0.2119)$

A1 ≈ 0.3102
r.t. 0.3102

Common error：把 *8* 部都當成條件概率 q，或把 *8* 部都當成 *0.2119*。

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11. Trapezoidal rule

a (undefined mark)

M1 $\frac{1}{2}[f(0)+f(4)+2(f(1)+f(2)+f(3))]$, $f(t)=\sqrt{1+e^{t/2}}$

A1 ≈ 8.0524
r.t. 8.0524

b (undefined mark)

M1 Let $u=\frac{t^2}{16}$, $du=\frac{t}{8}dt$: $\int_0^1 64e^u\,du$

A1 $64(e-1)\approx 109.9700$

c(i) (undefined mark)

M1 Change $=\int_0^4 8te^{t^2/16}dt-10\int_0^4\sqrt{1+e^{t/2}}dt$

M1 Use (a) and (b): $\approx 109.9700-10(8.0524)$

A1 ≈ 29.4463 m^3
r.t. 29.4463（用未四捨五入的 *(a)*）

c(ii) (undefined mark)

M1 $f'(t)=\frac{e^{t/2}}{4\sqrt{1+e^{t/2}}}$

A1 $f''(t)=\frac{e^{t/2}(2+e^{t/2})}{16(1+e^{t/2})^{3/2}}$
或等價式

M1 $f''(t)>0$, so the estimate in (a) is an over-estimate

A1 Melting is over-estimated, so the estimate in (c)(i) is an under-estimate

M1 Actual change $>29.4463>29$

A1 *Yes*
f.t.

Common error：忘記減去融雪量令高估／低估方向反轉。

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12. Applications of differentiation

a (undefined mark)

M1 $W = \frac{5000e^{-t/10} \cdot 3(1-e^{-t/5})}{1000} = 15\left(e^{-t/10} - e^{-3t/10}\right)$

M1 $\frac{dW}{dt} = 15\left(-\frac{1}{10}e^{-t/10} + \frac{3}{10}e^{-3t/10}\right)$

A1 $= \frac{3}{2}e^{-t/10}\left(3e^{-t/5} - 1\right)$

或等價式

b (undefined mark)

M1 $\frac{dW}{dt} = 0 \iff e^{-t/5} = \frac{1}{3}$; $\frac{dW}{dt} > 0$ for $t < 5\ln 3$, < 0 for $t > 5\ln 3$

1M for testing 包括在內

A1 $t = 5\ln 3$

c(i) (undefined mark)

M1 $15 \cdot 3^{-1/2}\left(1 - \frac{1}{3}\right)$

A1 $\frac{10\sqrt{3}}{3} \approx 5.7735$ (thousand kg)

c(ii) (undefined mark)

M1 $\frac{dW}{dt} > 0$ for \$05\ln 3, *so*W\$ increases then decreases

兩邊單調性都要

M1 $W(2) = 15\left(e^{-0.2} - e^{-0.6}\right) \approx 4.0488$

M1 $W(13) = 15\left(e^{-1.3} - e^{-3.9}\right) \approx 3.7843$

A1 Both exceed $\frac{5\sqrt{3}}{3} \approx 2.8868$, so W is greater than half of its greatest value throughout $2 \le t \le 13$ (11 months), and also for some $t < 2$; agree.

f.t.

Alternative：解 $W(t) = \frac{5\sqrt{3}}{3}$ 得 $t \approx 1.2267$，故首 *13* 個月內約有 *11.7733* 個月高於一半：全分。

Common error：只計 *[2, 13]* 剛好 *11* 個月就話「唔係多於 *11* 個月」，冇留意 $t < 2$ 都有一段高於一半。

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